

COMPSCI 1DM3 - Winter 2026

Discrete Mathematics for Computer Science

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Last Lecture:

- Sets: Unordered collection of distinct objects
- Intervals: For real numbers a and b such that $a \leq b$
 - $[a, b]$: All real numbers between a and b (including both)
 - $(a, b]$: All real numbers between a and b (including only b)
 - $[a, b)$: All real numbers between a and b (including only a)
 - (a, b) : All real numbers between a and b (excluding both)
- $A \subseteq B$ if and only if for any arbitrary $x \in A$, we have $x \in B$
- $A \subset B$ if and only if for any arbitrary $x \in A$, we have $x \in B$, and also $A \neq B$
- $A = B$ if and only if $A \subseteq B$ and $B \subseteq A$
- For any set A , its cardinality $|A|$ is the number of its members
- Power set $\mathcal{P}(A)$: The set of all subsets of A .

2.2 Set operations

Let A and B be two sets.

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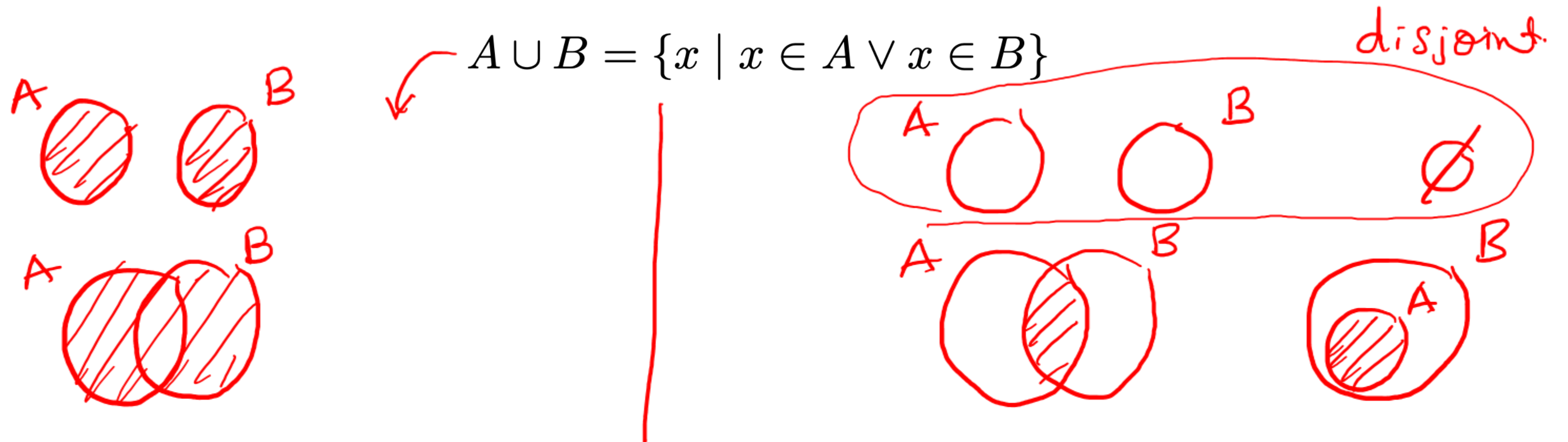
- Union: The union of A and B is defined as

$$A \cup B = \{x \mid x \in A \vee x \in B\}$$

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- Union: The union of A and B is defined as



- Intersection: The intersection of A and B is defined as

$$A \cap B = \{x \mid x \in A \wedge x \in B\}$$

Set operations

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- A and B are called disjoint if $A \cap B = \emptyset$, that is,

$$|\{x \mid x \in A \wedge x \in B\}| = 0$$

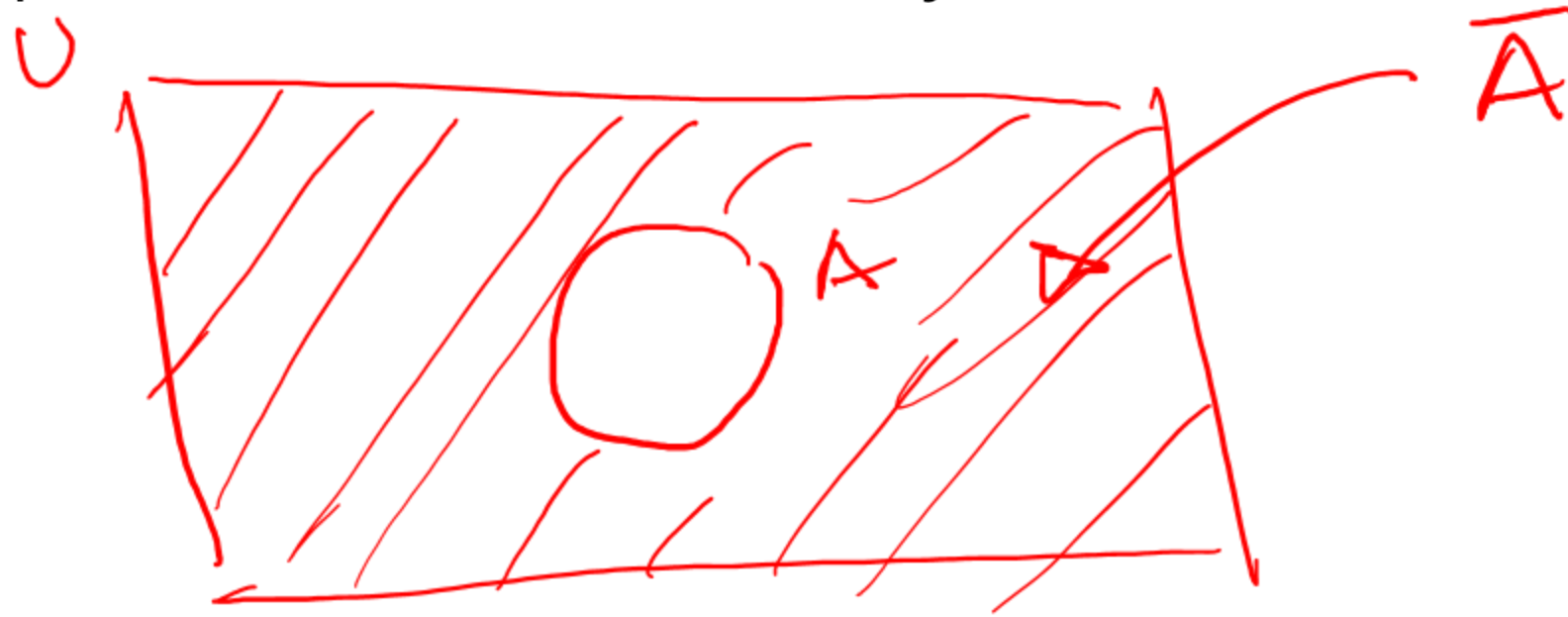
- Difference: The difference of A and B is defined as

$$\underline{A - B} = \{x \mid x \in A \wedge x \notin B\}$$



Sets

Let U be the possible domain for set's members. Then $U - A$ for a set A is called the complement of A , denoted by $\bar{A}$.



$$\begin{aligned} &A \cup (U - A) \\ &A \cup \bar{A} \\ &\quad \downarrow \\ &U \end{aligned}$$

Sets

Let U be the possible domain for set's members. Then $U - A$ for a set A is called the complement of A , denoted by $\bar{A}$.

$$\begin{array}{l} U = \mathbb{N} \\ \text{— even \#s} \\ \text{— } x \leq 11 \end{array} \left. \vphantom{\begin{array}{l} U = \mathbb{N} \\ \text{— even \#s} \\ \text{— } x \leq 11 \end{array}} \right\} \bar{A}$$
$$A = \{x \in \mathbb{N} \mid x > 11, x \text{ is odd}\}$$

Example

Find the complement of $A = \{x \text{ odd numbers} \mid x > 11\}$

such as line

$$\begin{array}{l} U = \text{odd numbers} \\ \text{— } x \leq 11 \end{array}$$

$$\bar{A} = \{1, 3, 5, 7, 9, 11\}$$

Sets

Symmetric difference: The set containing those elements in A or B , but not in both:

$$A \oplus B = \{x \mid x \in A \cup B \wedge x \notin A \cap B\}$$



Exercise

Show that $A - B = A \cap \bar{B}$.

$$\textcircled{1} A - B \subseteq A \cap \bar{B}$$

$$x \in A - B$$

$$\textcircled{x \in A}, x \notin B$$

$$\rightarrow \textcircled{x \in \bar{B}}$$

$$\therefore x \in A \cap \bar{B}$$

$$\textcircled{2} A \cap \bar{B} \subseteq A - B$$

$$x \in A \cap \bar{B}$$

$$x \in A \wedge x \in \bar{B}$$

$$x \in A \wedge x \notin B$$

$$x \in A - B$$

$\square$

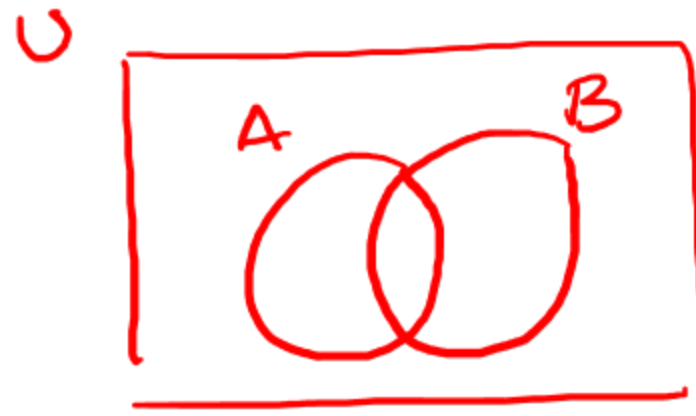
Set identities

- $A \cup U = U$, and $A \cup \emptyset = A$
- $A \cap U = A$, and $A \cap \emptyset = \emptyset$
- Commutative laws: $A \cup B = B \cup A$, and $A \cap B = B \cap A$
- Complementation law: $\overline{\overline{A}} = A$
- Associative laws:

$$A \cup (B \cup C) = (A \cup B) \cup C$$

$$A \cap (B \cap C) = (A \cap B) \cap C$$

Set identities



- De Morgan's laws:

$$\overline{(A \cap B)} = \overline{A} \cup \overline{B}$$

$$\overline{(A \cup B)} = \overline{A} \cap \overline{B}$$

- Distributive laws:

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

- $A \cup \overline{A} = U$, and $A \cap \overline{A} = \emptyset$

- Absorption laws:

$$A \cup (A \cap B) = A$$

$$A \cap (A \cup B) = A$$

How to use set identities

$$\boxed{B - A = B \cap \bar{A}}$$
$$\boxed{A - B = A \cap \bar{B}}$$

Show that

1. $A \cup (B - A) = A \cup B.$

$$A \cup (B \cap \bar{A}) = (A \cup B) \cap \underbrace{(A \cup \bar{A})}_{\mathcal{U}} = (A \cup B) \cap \mathcal{U} \\ = (A \cup B) \\ = \text{RHS} \quad \square$$

2. $(A \cap B) \cup (A - B) = A.$

$$\underbrace{(A \cap B)} \cup \underbrace{(A \cap \bar{B})} = A \cap \underbrace{(B \cup \bar{B})}_{\mathcal{U}} = \underline{A \cap \mathcal{U}} \\ = A = \text{RHS}$$

$\square$

Union and intersection: Generalization

Let $A_1, A_2, \dots, A_n$ be n sets. We define:

$$\bigcup_{i=1}^n A_i = A_1 \cup A_2 \cup \dots \cup A_n, \quad \bigcap_{i=1}^n A_i = A_1 \cap A_2 \cap \dots \cap A_n$$

Example

Let $A_i = \{1, 2, 3, \dots, i\}$. Then:

$$A_1 = \{1\}, \quad A_3 = \{1, 2, 3\}$$

$$A_2 = \{1, 2\}, \quad A_n = \{1, 2, \dots, n\}$$

- What is $\bigcup_{i=1}^n A_i$?

$$\rightarrow = A_1 \cup A_2 \cup \dots \cup A_n = \{1\} \cup \{1, 2\} \cup \dots \cup \{1, 2, \dots, n\} = \{1, 2, \dots, n\} = A_n$$

- What is $\bigcap_{i=1}^n A_i$?

$$= A_1 \cap A_2 \cap \dots \cap A_n = \{1\} \cap \{1, 2\} \cap \dots \cap \{1, 2, \dots, n\} = \{1\} = A_1$$

Multisets

- Recall that repetition is not allowed in sets. But in some cases, we want to emphasize repetition.
- How can we incorporate repetition into the definition of sets?

Multiset

Multiset is a generalization of sets in which repetition is allowed. For instance,

$$\{\underline{3.a}, \underline{2.b}, \underline{4.c}\}$$

describes a collection of three distinct objects a , b , c , where multiplicity of a is 3, multiplicity of b is 2, and multiplicity of c is 4.